MINOR PROJECT REPORT-2

Exploratory Data Analysis and Linear Regression

- Project title:*Exploratory Data Analysis and Linear Regression*
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- College:*VNR VJIET*
- Course/subject:Artifical intelligents
- Dataset name/Link:spotify-tracks-datasets-detailed.csv

1.INTRODUCTION

- **Project objective**

  To perform Exploratory Data Analysis (EDA) on the Spotify Tracks dataset to understand the characteristics of music tracks, identify patterns and relationships among different audio features, and analyze factors associated with track popularity. Linear regression is also used to examine the relationship between a selected audio feature and track popularity.

- **Dataset Name**

  Spotify tracks dataset

- **Dataset Source**

  Source: Kaggle
  Dataset: Spotify Tracks Dataset | Audio Features
  Source:
  https://www.kaggle.com/datasets/saichaitanyareddyai/spotify-tracks-dataset-audio-features/data

- **Short Description of the Dataset**

  The Spotify Tracks Dataset is a collection of 114,000 Spotify tracks covering 114 music genres. Each track contains information about the artist, album, track name, popularity, and several audio characteristics such as danceability, energy, loudness, acousticness, instrumentalness, liveness, valence, and tempo. The dataset can be used to explore music characteristics, analyze relationships between audio features, and study track popularity

- **Number of Rows and Columns(before cleaning)**

  Number of rows: 114,000
  Number of columns: 20

- **Important columns features**

Columns	**Description**
track_id	Unique identifier of the track
artists track_name	Name(s) of the performing artist(s) Name of the track

popularity	Popularity score of the track, from 0 to 100
duration_ms	Duration of the track in milliseconds
explicit	Indicates whether the track contains explicit content
danceability	Measures how suitable a track is for dancing
energy	Measures the intensity and activity of the track
loudness	Overall loudness of the track in decibels
acousticness	Measures how likely the track is to be acoustic
instrumentalness	Indicates how likely the track is to contain no vocals
liveness	Indicates the presence of a live audience
valence	Represents the musical positiveness of a track
tempo	Estimated tempo of the track in beats per minute
track_genre	Genre assigned to the track

2.DATA QUALITY & CLEANING

- **Missing values**

Columns	Missing values
Artists	1
Album_name	1
Track_name	1

Observation:

The dataset contained only three missing values, occurring in the artists, album_name, and track_name columns. Since the number of affected records was very small, the corresponding rows were removed. After cleaning, no missing values remained.

- **Duplicate values**

print("Number of duplicate rows:", df.duplicated().sum())

output:`np.int64(450)`

- **Datatype checking**

	0
track_id	object
artists	object
album_name	object
track_name	object
popularity	int64
duration_ms	int64
explicit	bool
danceability	float64

energy	float64
key	int64
loudness	float64
mode	int64
speechiness	float64
acousticness	float64
instrumentalness	float64
liveness	float64
valence	float64
tempo	float64
time_signature	int64
track_genre	object

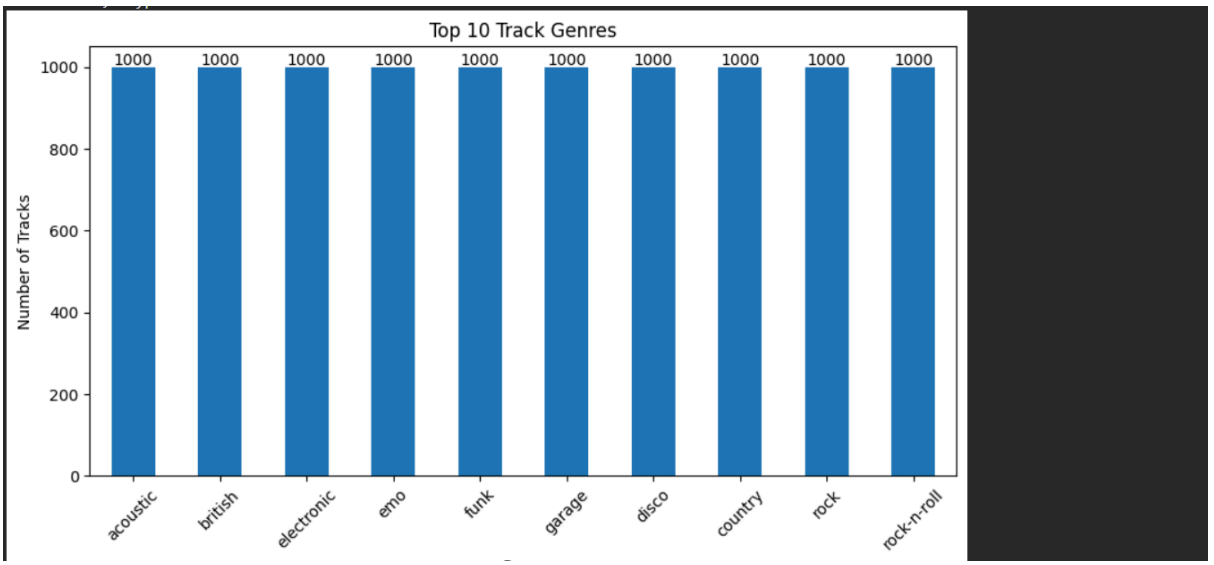
Observation:

The data types of the columns are appropriate for their respective values. Text-based columns are stored as object types, numerical features are stored as integer or float types, and the explicit column is stored as a Boolean type. Therefore, no major data type conversion was required.

- **Dataset after cleaning**
  113,997 rows x 20 columns
- **Overall Observation**
  The dataset required minimal cleaning. Three records containing missing values were removed, while the number of columns remained unchanged. The data types were found to be appropriate for the respective variables, and duplicate records were checked before proceeding with further analysis.

3.Categorical Analysis

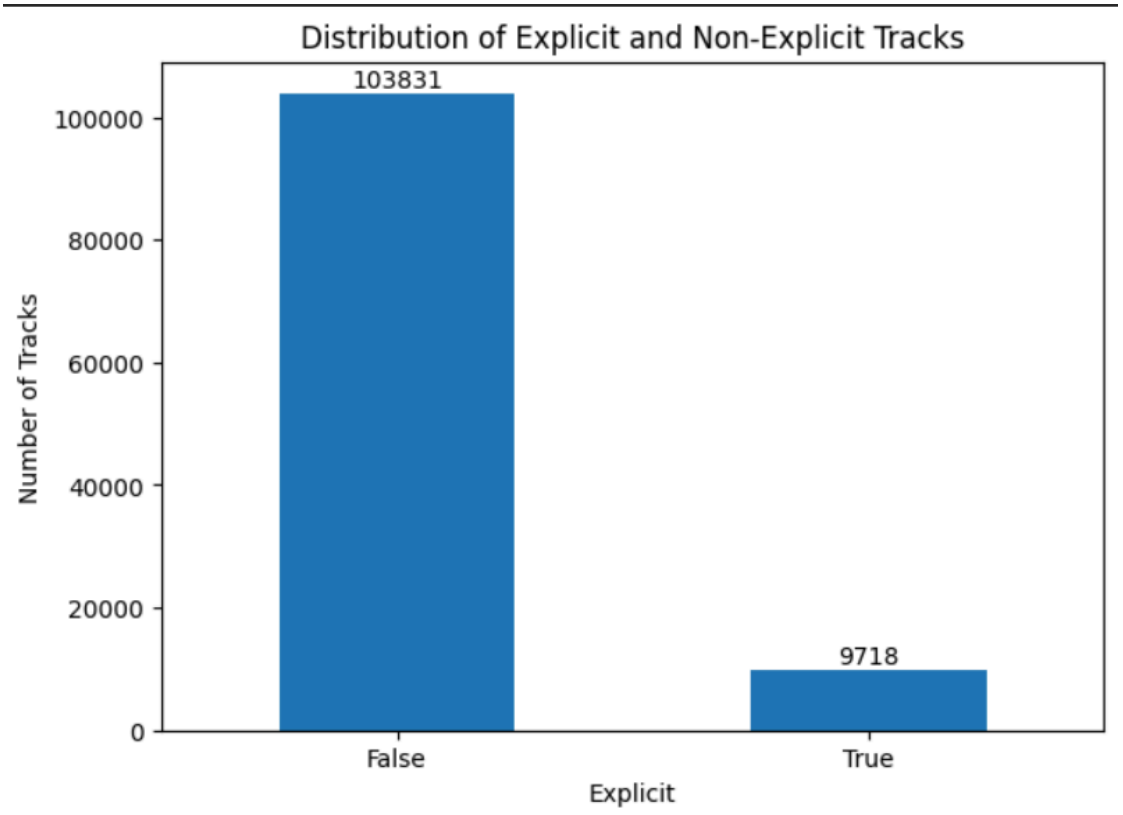
- ☐ **Genre chart**



Observation:

The chart shows that all the top 10 genres have an equal representation of 1,000 tracks each. Therefore, there is no difference in the number of tracks among these genres.

☐ **Explicit content**

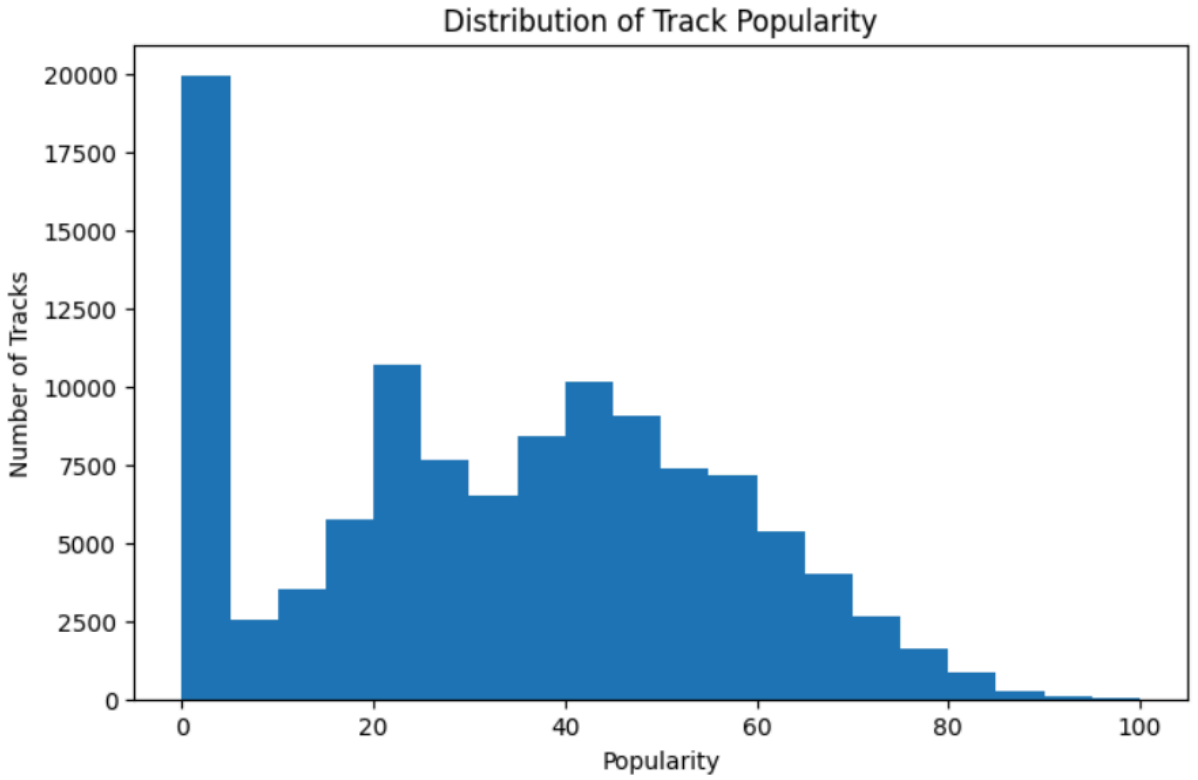


observation

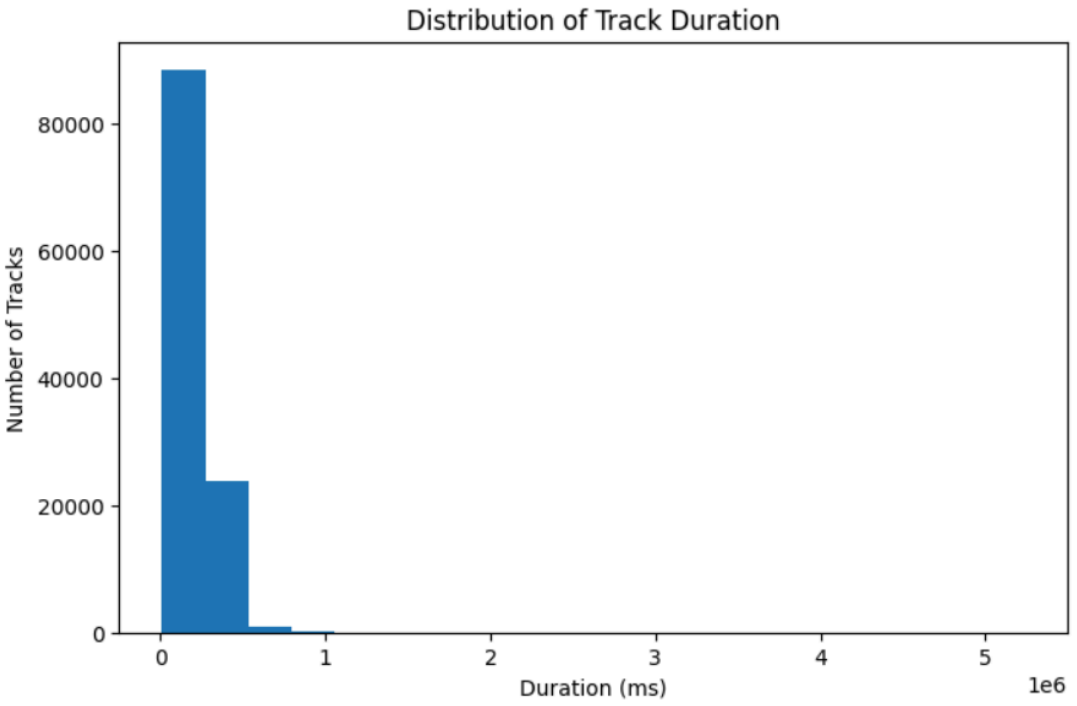
The dataset contains 103831 non-explicit tracks and 9,718 explicit tracks. Non-explicit tracks therefore form the majority of the dataset.

4.Univariate Analysis

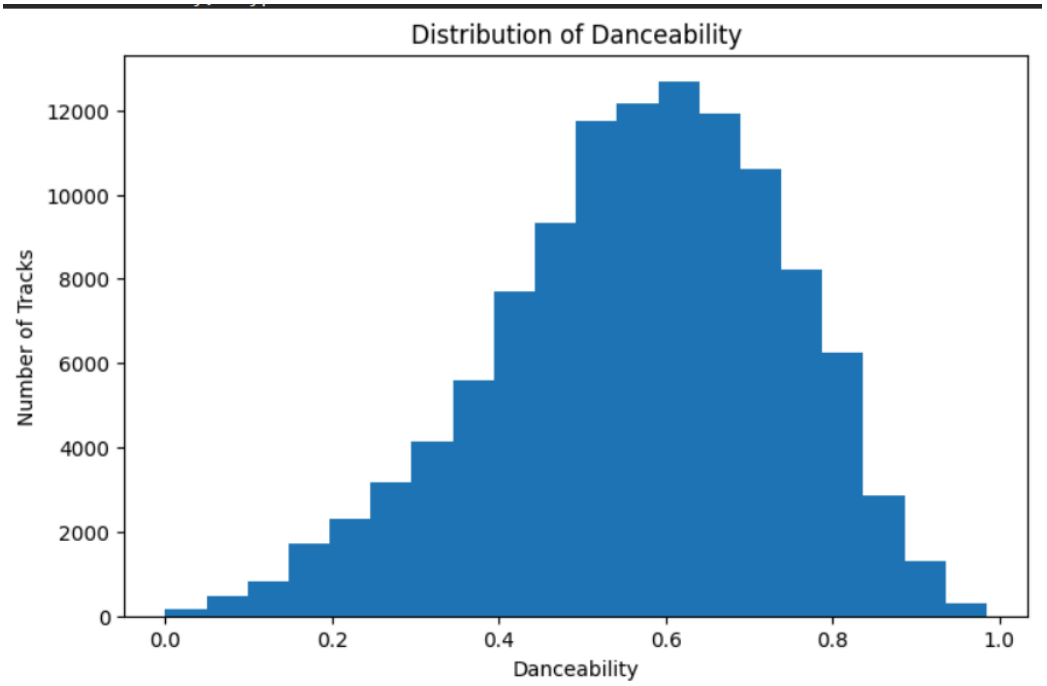
☐ Popularity



☐ Duration

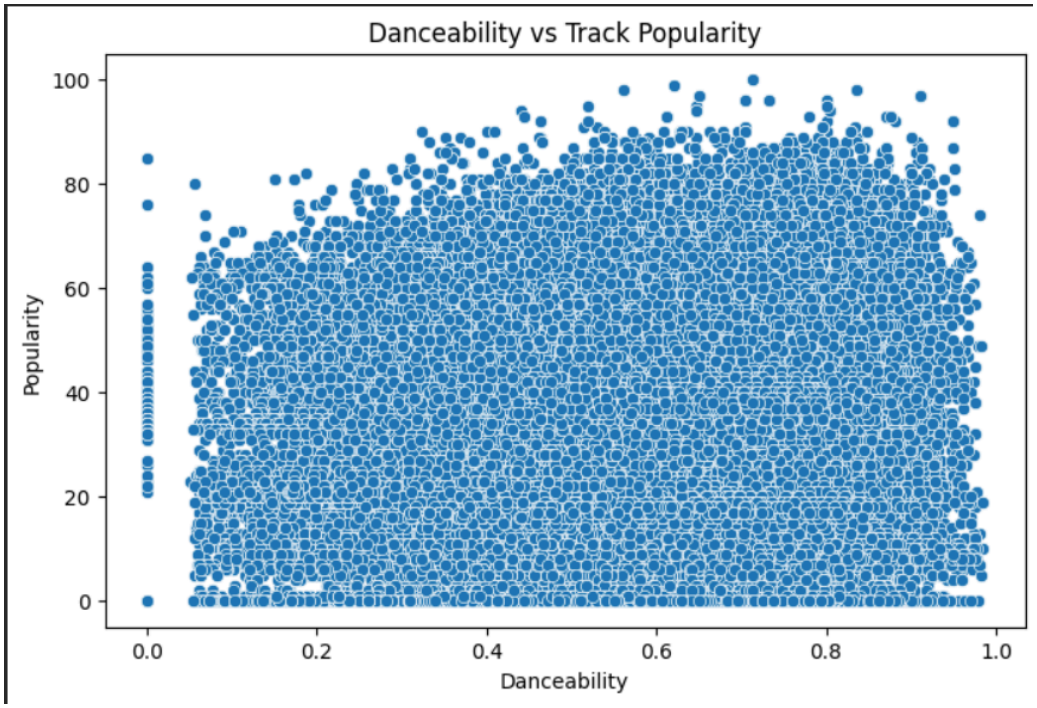


☐ Danceability



5.Bivariate & Multivariate Analysis

- ☐ Bivariate chart

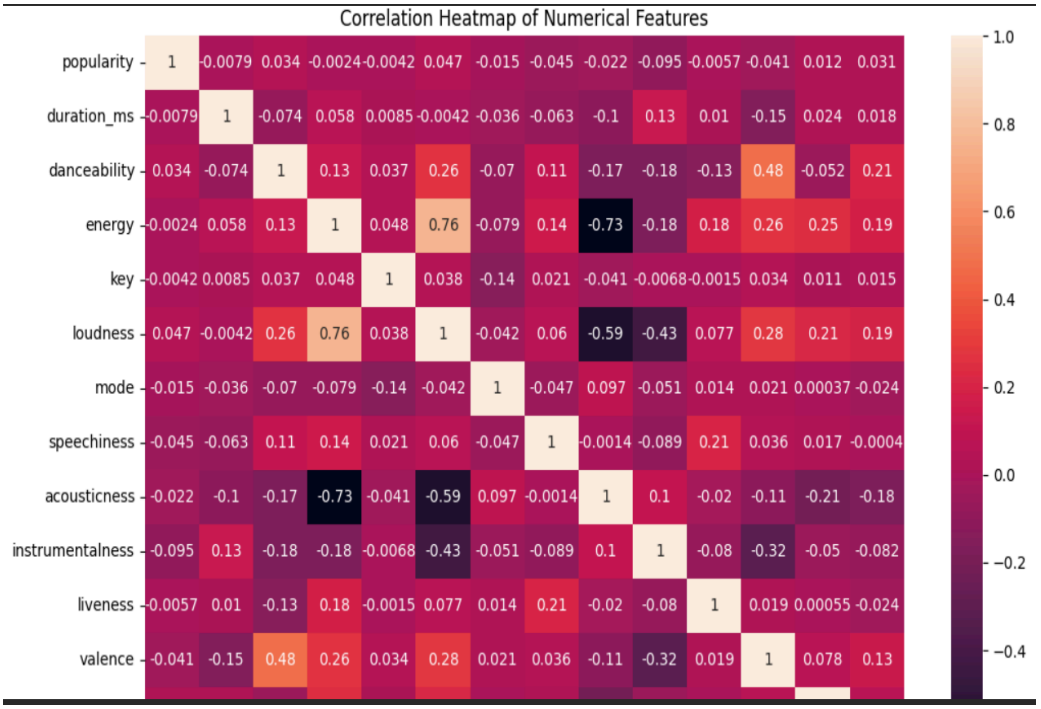


Observation: The scatter plot indicates that the relationship between danceability and popularity appears weak.

- ☐ Correlation Analysis

	popularity	duration_ms	danceability	energy	key	loudness	mode	speechiness	acousticness	instrumentalness	liveness	valence	tempo	time
popularity	1.000000	-0.007928	0.034407	-0.002447	-0.004167	0.047368	-0.015009	-0.045463	-0.022356	-0.094718	-0.005668	-0.041097	0.012187	
duration_ms	-0.007928	1.000000	-0.074117	0.058160	0.008519	-0.004233	-0.035961	-0.063194	-0.103676	0.125301	0.010199	-0.154795	0.024312	
danceability	0.034407	-0.074117	1.000000	0.131694	0.036546	0.256559	-0.069783	0.108236	-0.169181	-0.183974	-0.131651	0.476755	-0.051517	
energy	-0.002447	0.058160	0.131694	1.000000	0.048047	0.760624	-0.079331	0.141976	-0.732747	-0.179967	0.184810	0.258451	0.247361	
key	-0.004167	0.008519	0.036546	0.048047	1.000000	0.038359	-0.135871	0.020820	-0.041124	-0.006819	-0.001518	0.034233	0.010903	
loudness	0.047368	-0.004233	0.256559	0.760624	0.038359	1.000000	-0.042074	0.060088	-0.588111	-0.432109	0.076792	0.279428	0.212181	
mode	-0.015009	-0.035961	-0.069783	-0.079331	-0.135871	-0.042074	1.000000	-0.046773	0.096548	-0.050787	0.013900	0.021411	0.000371	
speechiness	-0.045463	-0.063194	0.108236	0.141976	0.020820	0.060088	-0.046773	1.000000	-0.001378	-0.089214	0.205092	0.036393	0.017167	
acousticness	-0.022356	-0.103676	-0.169181	-0.732747	-0.041124	-0.588111	0.096548	-0.001378	1.000000	0.102134	-0.020313	-0.106240	-0.207620	
instrumentalness	-0.094718	0.125301	-0.183974	-0.179967	-0.006819	-0.432109	-0.050787	-0.089214	0.102134	1.000000	-0.079847	-0.324314	-0.049939	
liveness	-0.005668	0.010199	-0.131651	0.184810	-0.001518	0.076792	0.013900	0.205092	-0.020313	-0.079847	1.000000	0.019339	0.000547	
valence	-0.041097	-0.154795	0.476755	0.258451	0.034233	0.279428	0.021411	0.036393	-0.106240	-0.324314	0.019339	1.000000	0.077640	
tempo	0.012187	0.024312	-0.051517	0.247361	0.010903	0.212181	0.000371	0.017167	-0.207620	-0.049939	0.000547	0.077640	1.000000	
time_signature	0.030567	0.018351	0.207370	0.186671	0.015122	0.191648	-0.024365	-0.000397	-0.175621	-0.081951	-0.023902	0.133790	0.066900	

☐ Heat map

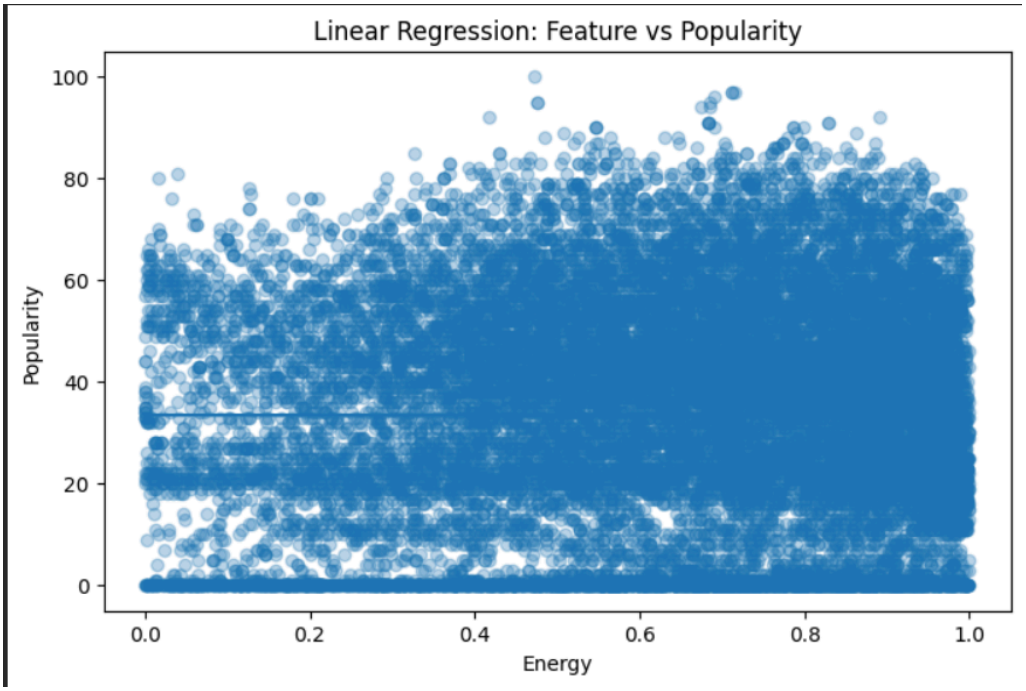


6.Key Insights & Linear Regression

1. The dataset contains a large number of non-explicit tracks compared with explicit tracks.
2. Track popularity is unevenly distributed, with a large number of tracks having very low popularity.
3. Track duration is strongly right-skewed, with most tracks having shorter durations and a small number of tracks having very large duration values.
4. Danceability values are concentrated mainly around the moderate-to-high range, particularly between approximately 0.5 and 0.7.
5. Energy and loudness show a relatively strong positive relationship, while danceability has a relatively weak relationship with popularity.

6. The genre distribution is balanced, with each genre containing approximately 1000 tracks.

- **Linear regression**
  Target variable:popularity
  Selected feature:Energy



Model Evaluation

```
MAE: 18.969042871880305
MSE: 501.6166187641697
RMSE: 22.396799297314107
R² Score: -5.688770317924252e-06
```

7.Limitation

The analysis is based on the available Spotify track dataset and its recorded audio features. The analysis identifies relationships and patterns in the data but does not establish cause-and-effect relationships. Some unusual values may also require further investigation to determine whether they represent genuine tracks or data irregularities.

8.Conclusion

This project explored the Spotify Tracks dataset through data cleaning, descriptive statistics, univariate analysis, bivariate analysis, multivariate analysis, and visualization. The analysis helped identify patterns in track popularity, duration, danceability, explicit content, and other audio features. Correlation analysis was used to examine relationships between numerical variables, and linear regression was performed to study the relationship between the selected feature and popularity. Overall, the project provided an understanding of the structure, distribution, relationships, and patterns present in the dataset.